



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 6

1. Title: Use string functions and regex for email validation, data extraction, and text analysis.

2. Software/Tools Required:

1. Visual Studio Code (VS Code)
2. Google Chrome
3. HTML5
4. JavaScript
5. CSS

3. Theory: String functions in JavaScript are used to manipulate and analyze text. Common methods such as `length`, `toUpperCase()`, `toLowerCase()`, `trim()`, `split()`, `replace()`, and `includes()` help process string data efficiently. **Regular Expressions (Regex)** are patterns used to search, validate, and extract text. They are commonly used for validating email addresses, finding specific words or numbers, and performing text analysis. This experiment demonstrates how string methods and Regex can be combined to validate user input, extract useful information, and analyze textual data effectively.



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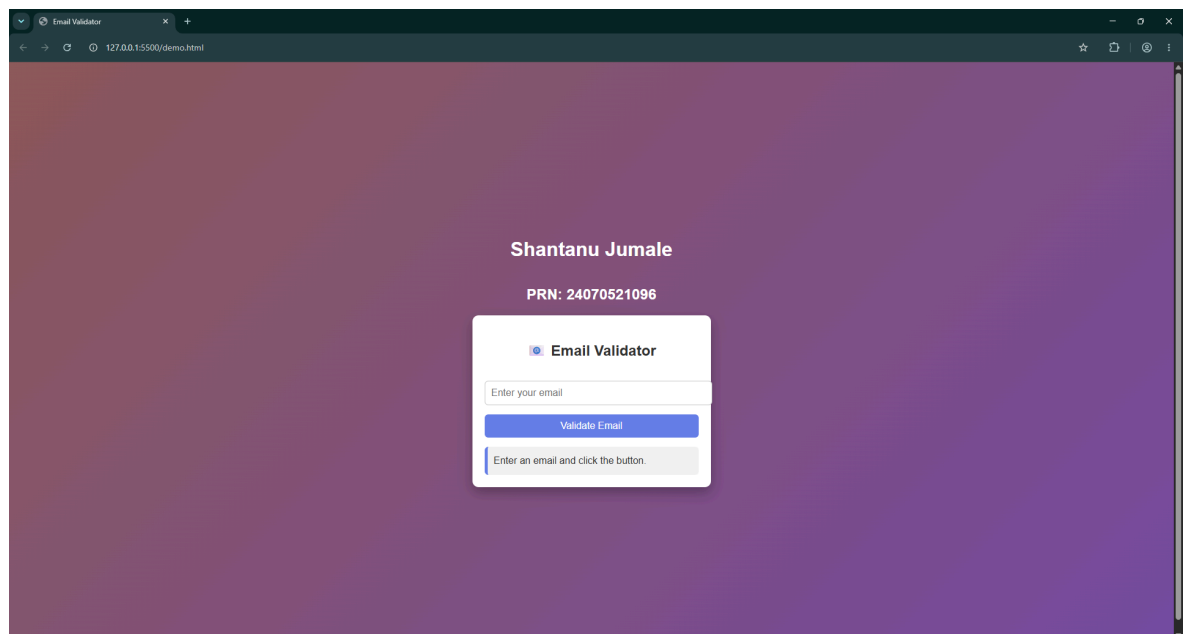
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4. Output:



5. Result/Conclusion:

The program was successfully implemented using JavaScript string functions and regular expressions. It accurately validates email addresses, extracts required data, and performs basic text analysis. The application demonstrates efficient text processing and input validation techniques. This experiment enhances understanding of JavaScript string manipulation and the practical use of Regex in web development.